0.0.1 p-norm in $\mathbb{R}^n$

Definition 0.0.1. Consider $\mathbb{R}^n$ and $p \in [1, \infty]$. Define *p-norm* on $\mathbb{R}^n$ as:

$$\|\cdot\|_p : \mathbb{R}^n \to \mathbb{R} : (x_i)_{i=1}^n \mapsto \left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}}$$

This induces a Mectric on $\mathbb{R}^n$ as:

$$d_p : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R} : ((x_i)_{i=1}^n, (y_i)_{i=1}^n) \mapsto \|(x_i - y_i)_{i=1}^n\|_p$$

Note: d_p satisfies the Axioms of Metric, as it is induced by a norm.

Lemma 0.0.2. *Young's inequality*
Let $u, v > 0$, and $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$.
Then,

$$uv \leq \frac{1}{p}u^p + \frac{1}{q}v^q$$

Proof. Since $f(x) = \log x$ is concave, we obtain

$$\forall \lambda \in [0, 1], \;\; \lambda f(x) + (1 - \lambda)f(y) \leq f(\lambda x + (1 - \lambda)y)$$

thus,

$$\log\left(\frac{1}{p}u^p + \frac{1}{q}v^q\right) \geq \frac{1}{p}\log(u^p) + \frac{1}{q}\log(v^q) = \log(uv)$$

Since $\exp(x)$ is increase function, we have

$$\exp\left(\log\left(\frac{1}{p}u^p + \frac{1}{q}v^q\right)\right) \geq \exp(\log(uv))$$

That is,

$$uv \leq \frac{1}{p}u^p + \frac{1}{q}v^q$$

□

Lemma 0.0.3. *Holder's inequality*
Let $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ be given, and $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then,

$$\|\mathbf{xy}\|_1 = \sum_{i=1}^n |x_i y_i| \leq \left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} \cdot \left(\sum_{i=1}^n |y_i|^q\right)^{\frac{1}{q}} = \|\mathbf{x}\|_p \|\mathbf{y}\|_q$$

Proof. Since Young's inequality, for each $i \in \{1, 2, \ldots, n\}$, it satisfies that

$$\frac{|x_i|}{\|\mathbf{x}\|_p} \cdot \frac{|y_i|}{\|\mathbf{y}\|_p} \leq \frac{1}{p} \cdot \frac{|x_i|^p}{\|\mathbf{x}\|_p^p} + \frac{1}{q} \cdot \frac{|y_i|^q}{\|\mathbf{y}\|_q^q}$$

Sum all terms for $i = 1, 2, \ldots, n$, then

$$\frac{1}{\|\mathbf{x}\|_p \|\mathbf{y}\|_q} \cdot \sum_{i=1}^n |x_i y_i| \leq \frac{1}{p} + \frac{1}{q} = 1$$

In conclusion,

$$\sum_{i=1}^n |x_i y_i| \leq \left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} \cdot \left(\sum_{i=1}^n |y_i|^q\right)^{\frac{1}{q}}$$

□

> **Theorem 0.0.4.** *Minkowski inequality*
> Given complex-valued tuples $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$,
>
> $$\|\mathbf{x}+\mathbf{y}\|_p = \left[\sum_{i=1}^{n} |x_i + y_i|^p\right]^{\frac{1}{p}} \le \left[\sum_{i=1}^{n} |x_i|^p\right]^{\frac{1}{p}} + \left[\sum_{i=1}^{n} |y_i|^p\right]^{\frac{1}{p}} = \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$$

Proof. Denote

$$|x_i + y_i|^p = |x_i + y_i| \cdot |x_i + y_i|^{p-1}$$

Then,

$$\begin{aligned}
\sum_{i=1}^{n} |x_i + y_i|^p &= \sum_{i=1}^{n} |x_i + y_i| \cdot |x_i + y_i|^{p-1} \\
&= \sum_{i=1}^{n} (|x_i| + |y_i|) \cdot |x_i + y_i|^{p-1} \\
&= \sum_{i=1}^{n} |x_i| \cdot |x_i + y_i|^{p-1} + \sum_{i=1}^{n} |y_i| \cdot |x_i + y_i|^{p-1} \\
&\overset{\text{Hölder}}{\le} \left[\sum_{i=1}^{n} |x_i|^p\right]^{\frac{1}{p}} \cdot \left[\sum_{i=1}^{n} |x_i + y_i|^p\right]^{\frac{p-1}{p}} + \left[\sum_{i=1}^{n} |y_i|^p\right]^{\frac{1}{p}} \cdot \left[\sum_{i=1}^{n} |x_i + y_i|^p\right]^{\frac{p-1}{p}} \\
&= \left[\left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{\frac{1}{p}}\right] \cdot \left[\sum_{i=1}^{n} |x_i + y_i|^p\right]^{\frac{p-1}{p}}
\end{aligned}$$

Now, Divide each side by $\left[\sum_{i=1}^{n} |x_i + y_i|^p\right]^{\frac{p-1}{p}}$, then we obtain

$$\left[\sum_{i=1}^{n} |x_i + y_i|^p\right]^{1-\frac{p-1}{p}} = \left[\sum_{i=1}^{n} |x_i + y_i|^p\right]^{\frac{1}{p}} \le \left[\left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{\frac{1}{p}}\right]$$

□

0.0.2 Every p-Norm Spaces are Equivalent

> **Theorem 0.0.5.** Let d_{p_1}, d_{p_2} are p-norm on $\mathbb{R}^n$ with $1 \le p_1 < p_2 \le \infty$. Then,
>
> $$\exists C > 0 \ \text{ s.t } \ \forall x, y \in \mathbb{R}^n, \ \ d_{p_2}(x,y) \le d_{p_1}(x,y) \le C d_{p_2}(x,y)$$
>
> Specifically, $C = n^{\frac{1}{p_1} - \frac{1}{p_2}}$.

Proof. Let $p_1 < p_2$.
~~For~~ To show that first-inequality,

$$1 = \sum_{i=1}^{n} \left[\frac{|x_i - y_i|}{\left[\sum_{i=1}^n |x_i - y_i|^{p_2}\right]^{\frac{1}{p_2}}} \right]^{p_2} \le \sum_{i=1}^{n} \left[\frac{|x_i - y_i|}{\left[\sum_{i=1}^n |x_i - y_i|^{p_2}\right]^{\frac{1}{p_2}}} \right]^{p_1} = \frac{\sum_{i=1}^n |x_i - y_i|^{p_1}}{\left[\sum_{i=1}^n |x_i - y_i|^{p_2}\right]^{\frac{p_1}{p_2}}} = \left[\frac{\left[\sum_{i=1}^n |x_i - y_i|^{p_1}\right]^{\frac{1}{p_1}}}{\left[\sum_{i=1}^n |x_i - y_i|^{p_2}\right]^{\frac{1}{p_2}}} \right]^{p_1}$$

Thus, we obtain that:

$$1 \le \left[\frac{\left[\sum_{i=1}^n |x_i - y_i|^{p_1}\right]^{\frac{1}{p_1}}}{\left[\sum_{i=1}^n |x_i - y_i|^{p_2}\right]^{\frac{1}{p_2}}} \right]^{p_1} \iff 1 \le \frac{\left[\sum_{i=1}^n |x_i - y_i|^{p_1}\right]^{\frac{1}{p_1}}}{\left[\sum_{i=1}^n |x_i - y_i|^{p_2}\right]^{\frac{1}{p_2}}} \iff \left[\sum_{i=1}^n |x_i - y_i|^{p_2} \right]^{\frac{1}{p_2}} \le \left[\sum_{i=1}^n |x_i - y_i|^{p_1} \right]^{\frac{1}{p_1}}$$

To show that second-inequality, using Hölder's inequality.

$$\begin{aligned}
(d_{p_1}(x,y))^{p_1} &= \sum_{i=1}^n |x_i - y_i|^{p_1} = \sum_{i=1}^n |x_i - y_i|^{p_1} \cdot 1 \\
&\overset{\text{Hölder}}{\le} \left[\sum_{i=1}^n \left(|x_i - y_i|^{p_1 \cdot \frac{p_2}{p_1}} \right) \right]^{\frac{p_1}{p_2}} \cdot \left[\sum_{i=1}^n 1^{\frac{p_2}{p_2 - p_1}} \right]^{1 - \frac{p_1}{p_2}} = \left[\sum_{i=1}^n \left(|x_i - y_i|^{p_2} \right) \right]^{\frac{p_1}{p_2}} \cdot n^{1 - \frac{p_1}{p_2}}
\end{aligned}$$

Take the $\frac{1}{p_1}$-th power to both sides, then

$$d_{p_1}(x,y) \le \left[\sum_{i=1}^n \left(|x_i - y_i|^{p_2} \right) \right]^{\frac{1}{p_2}} \cdot n^{\frac{1}{p_1} - \frac{1}{p_2}} = n^{\frac{1}{p_1} - \frac{1}{p_2}} \cdot d_{p_2}(x,y)$$

□

> **Corollary 0.0.6.** Let Metrics on $\mathbb{R}^n$ induced by p-norms $d_{p_1}, d_{p_2} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ be given.
> Then, the Topologies generated by this Metrics satisfy
>
> $$\mathcal{T}_{d_{p_1}} = \mathcal{T}_{d_{p_2}}$$